Classical ML Refresher Through a Quantum Lens

EVA: Quantum Machine Learning – Lecture 4

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Roadmap

1. Linear models in matrix form
2. Feature maps and kernels
3. Bridge to quantum state embeddings

Linear models and decision functions

Supervised learning in matrix form

With data matrix $X \in \mathbb{R}^{N \times d}$, parameters $w \in \mathbb{R}^d$, and bias $b \in \mathbb{R}$:

$$z = Xw + b1, \qquad \hat{y} = g(z).$$

Typical choices:

- regression: $g(z) = z$
- binary classification: $g(z) = \sigma(z) = \frac{1}{1+e^{-z}}$

`lecture_04_demo.ipynb`, Demo 1: matrix-form linear/logistic baseline.

Logistic regression objective

For labels $y_i \in \{0, 1\}$ and probabilities $p_i = \sigma(w^\top x_i + b)$, the negative log-likelihood is

$$\mathcal{L}(w, b) = -\sum_{i=1}^{N} \left[y_i \log p_i + (1 - y_i) \log(1 - p_i) \right].$$

The decision boundary is still linear in the original feature space:

$$w^\top x + b = 0.$$

Question pause 1

Discuss in pairs:

- Why can logistic regression still fail on XOR?
- What information is missing in the original 2D features?

Kernels and feature maps

Classical ML: powerful but feature-dependent

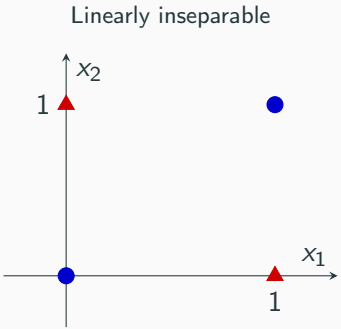
The classical ML toolbox is mature and effective:

- kernel methods, deep networks, ensemble methods
- strong performance across classification and regression tasks

Central issue: feature representation.

- the right feature map can make a hard problem easy
- the wrong feature map can make a simple problem hard

The XOR problem



No hyperplane in $\mathbb{R}^2$ separates the classes.

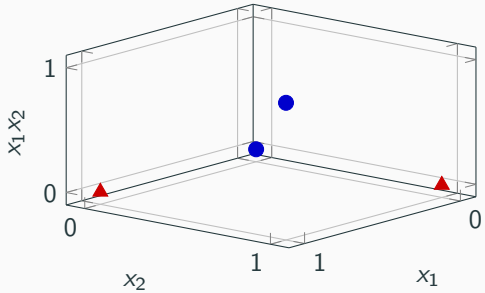
Classical fix: nonlinear lifting

$$\phi(x_1, x_2) = (x_1, x_2, x_1 x_2)$$

In $\mathbb{R}^3$, XOR becomes linearly separable.

XOR after lifting to 3D

$$\phi(x_1, x_2) = (x_1, x_2, x_1 x_2)$$



One separating hyperplane:

$$s = x_1 + x_2 - 2x_3 - \frac{1}{2}$$

- $(0, 0, 0)$ and $(1, 1, 1)$: $s = -0.5$
- $(0, 1, 0)$ and $(1, 0, 0)$: $s = +0.5$

`lecture_04_demo.ipynb`, Demo 2: XOR lifting visualization.

Kernel trick on XOR

Inputs:

$$x^{(1)} = (0,0),\ x^{(2)} = (0,1),\ x^{(3)} = (1,0),\ x^{(4)} = (1,1)$$

Polynomial kernel:

$$K(x,y) = (x^\top y + 1)^2$$

The Gram matrix becomes:

$$K = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 4 & 1 & 4 \\ 1 & 1 & 4 & 4 \\ 1 & 4 & 4 & 9 \end{bmatrix}$$

`lecture_04_demo.ipynb`, Demo 3: kernel matrix and kernel-SVM comparison.

Kernel families at a glance

Kernel	Formula	Typical behavior
Linear	$x^\top y$	simple, robust baseline
Polynomial	$(\gamma x^\top y + c)^d$	interactions up to degree d
RBF	$\exp\left(-\gamma \lVert x - y \rVert^2\right)$	local similarity, flexible boundaries

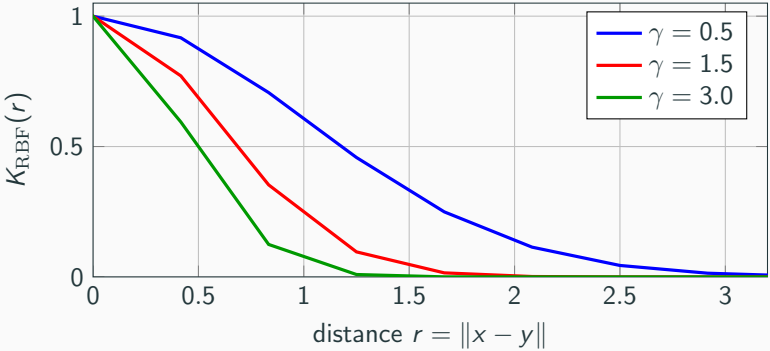
No kernel is universally best. Always compare under the same split and hyperparameter tuning protocol.

`lecture_04_demo.ipynb`, Demo 4: linear vs RBF on Iris subset.

Visualization: how RBF γ changes locality

RBF kernel:

$$K_{\mathrm{RBF}}(r) = \exp\left(-\gamma r^2\right), \qquad r = \|x - y\|.$$



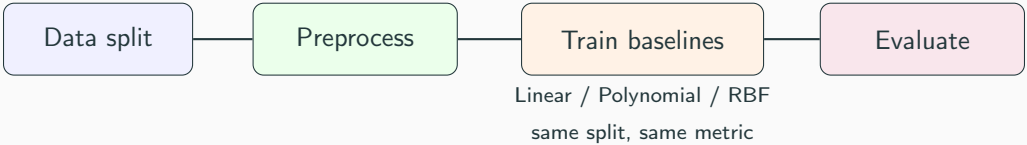
Larger γ makes similarity decay faster (more local decision behavior).

Question pause 2

Open questions for class:

- If RBF is very flexible, why not always use it?
- What can go wrong when γ is too large or too small?
- What would be a fair baseline protocol before claiming improvement?

Visualization: baseline comparison workflow



The same protocol is used before introducing quantum feature maps.

Classical-to-quantum bridge

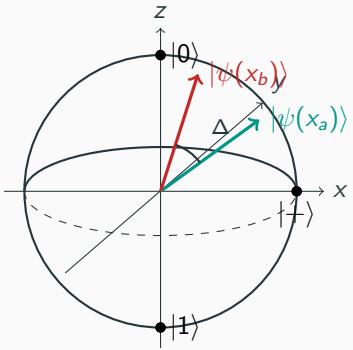
Classical kernel logic:

$$K(x, y) = \langle \phi(x), \phi(y) \rangle$$

Quantum analogue:

$$|\phi(x)\rangle = U_\phi(x) \, |0\rangle^{\otimes n} \,, \qquad K_Q(x, y) = |\langle \phi(x) | \phi(y) \rangle|^2 \,.$$

Bloch-sphere geometry (one-qubit intuition)



Feature encoding can be viewed geometrically as mapping data points to quantum states on the Bloch sphere.

Smaller angular separation Δ means larger overlap:

$$|\langle\psi(x_a)|\psi(x_b)\rangle|^2 = \cos^2\left(\frac{\Delta}{2}\right)$$

for pure one-qubit states.

Question pause 3

Quick check:

- In classical kernels, similarity is $\langle \phi(x), \phi(y) \rangle$.
- In quantum kernels, similarity is $|\langle \phi(x) | \phi(y) \rangle|^2$.

What is gained and what is lost by squaring the overlap magnitude?

Bridge to Class 5

In Lecture 5, we turn this intuition into executable quantum feature maps and overlap-based kernels.